Z.H. College of Engineering & Technology, Aligarh Muslim University Aligarh
Course Structure: B.Tech. VLSI
Valid for students admitted in year 2024 onwards

Semester 3:

S.No.	Course Catag.	Crs No.	Course title	Contact Periods			Credits	Marks			Total	Pre-Requisite Courses
				L	T	P		Course-Work	Mid-Sem	End-Sem		
1	BS	AMS2512	Higher Mathematics	3	1	0	4	15	25	60	100	
2	BS	AMS2522	Numerical Methods and Optimization	3	0	0	3	15	25	60	100	
3	PC	ELC2112	Electronic Devices & Circuits	3	1	0	4	15	25	60	100	ELA1112
4	PC	ELC2122	Circuit Theory	3	1	0	4	15	25	60	100	
5	PC	ELC2312	Logic Circuits	3	1	0	4	15	25	60	100	
6	PC	ELC2412	Signals and Systems	3	1	0	4	15	25	60	100	AMS1122
7	PC	ELC2942	Electronic Circuits Lab	0	0	3	1.5	60		40	100	
8	AU		Audit Course-2				0					

TOTAL CREDITS: 24.5

Semester 4:

S.No.	Course Catag.	Crs No.	Course title	Contact Periods			Credits	Marks			Total	Pre-Requisite Courses
				L	T	P		Course-Work	Mid-Sem	End-Sem		
1	ESA	COA2602	Data Structures and Programming	2	1	0	3	15	25	60	100	COA1912
2	PC	ELC2142	Physics of Semiconductor Devices	3	1	0	4	15	25	60	100	EEA1112, ELC2112
3	ESA	ELA2432	Representation and Analysis of random Signals	3	0	0	3	15	25	60	100	AMS2512
4	PC	ELC2512	Electromagnetics	3	1	0	4	15	25	60	100	
5	PC	ELC2132	Digital Electronics	3	0	0	3	15	25	60	100	ELC2312
6	PC	ELC2152	Analog Electronics	3	1	0	4	15	25	60	100	ELC2112, ELC2122
7	PC	ELC2952	Digital Electronics Lab	0	0	3	1.5	60		40	100	
8	PC	ELC2962	Circuit Simulation Lab	0	0	3	1.5	60		40	100	

TOTAL CREDITS: 24

Z.H. College of Engineering & Technology, Aligarh Muslim University Aligarh

Course Structure: B.Tech. VLSI

Semester 5:

S.No.	Course Catag.	Crs No.	Course title	Contact Periods			Credits	Marks			Total	Pre-Requisite Courses
				L	T	P		Course-Work	Mid-Sem	End-Sem		
1	PC	ELC3212	Control Systems	3	1	0	4	15	25	60	100	ELC2122, ELC2412
2	PC	ELC3332	Embedded System	3	1	0	4	15	25	60	100	ELC2132
3	PC	ELC3442	Principle of Communication Systems	3	1	0	4	15	25	60	100	ELC2412
4	PC	ELC3632	Analog IC Design	3	0	0	3	15	25	60	100	ELC2152
5	PC	ELC3642	DIgital IC Design	3	0	0	3	15	25	60	100	ELC2132, ELC2152
6	PC	ELC3622	Digital System Design	3	0	0	3	15	25	60	100	ELC2312
7	PC	ELC3972	Digital System Design Lab	0	1	2	2	60		40	100	
8	PC	ELC3912	Circuit Design Lab	0	1	2	2	60		40	100	

TOTAL CREDITS: 25

Semester 6:

S.No.	Course Catag.	Crs No.	Course title	Contact Periods			Credits	Marks			Total	Pre-Requisite Courses
				L	T	P		Course-Work	Mid-Sem	End-Sem		
1	PE		Program Elective - 1	3	0	0	3				100	
2	PC	ELC3122	Analog Signal Processing	3	1	0	4	15	25	60	100	ELC2152
3	HM	MEH3452	Engineering Economy & Management	3	0	0	3	15	25	60	100	
4	PC	ELC3652	RF IC Design	3	0	0	3	15	25	60	100	ELC3632
5	PC	ELC3432	Digital Signal Processing	3	1	0	4	15	25	60	100	ELC2412
6	PSI	ELP3992	Mini Project	0	0	6	3	60		40	100	
7	PC	ELC3982	DSP Systems Lab	0	0	3	1.5	60		40	100	
8	PC	ELC3962	Embedded System Lab	0	1	2	2	60		40	100	
9	AU		Audit Course-3				0					

TOTAL CREDITS: 23.5

Z.H. College of Engineering & Technology, Aligarh Muslim University Aligarh

Course Structure: B.Tech. VLSI

Semester 7:

S.No.	Course Catag.	Crs No.	Course title	Contact Periods			Credits	Marks			Total	Pre-Requisite Courses
				L	T	P		Course-Work	Mid-Sem	End-Sem		
1	OE		Open Elective - 1	3	0	0	3				100	
2	ESA	ELA4342	Computer Architecture	3	0	0	3	15	25	60	100	ELC3332, ELC3622
3	HM		Elective Course under HM Category	3	0	0	3	15	25	60	100	
3	PC	ELC4612	IC Fabrication & Technology	3	1	0	4	15	25	60	100	ELC2142, ELC2132, ELC2152
5	PE		Program Elective - 2	3	0	0	3				100	
6	PE		Program Elective - 3	3	0	0	3				100	
7	PC	ELC4932	IC Design Lab	0	1	2	2	60		40	100	
8	PSI	ELP4982	Project Phase - 1	0	0	8	4	60		40	100	

TOTAL CREDITS: 25

Semester 8:

S.No.	Course Catag.	Crs No.	Course title	Contact Periods			Credits	Marks			Total	Pre-Requisite Courses
				L	T	P		Course-Work	Mid-Sem	End-Sem		
1	PSI	ELP4902	Industrial Training/Internship				2			100	100	
2	OE		Open Elective - 2	3	0	0	3				100	
3	PE		Program Elective - 4	3	0	0	3				100	
4	PE		Program Elective - 5	3	0	0	3				100	
5	PSI	ELP4992	Project Phase - 2	0	0	8	4	60		40	100	
6											100	
7											100	
8											100	
9											100	
10											100	

TOTAL CREDITS: 15

SUMMARY:

Catagory Code	Category Name	Credits Provided	Required		Provided in FY	To be provided in II Yr onwards	
			Min	Max		Min	Max
PC	Programme Core	81	60	100	0	60	100
PE	Programme Elective	15	15	32	0	15	32
BS	Basic Sciences	7	25	30	19	6	11
ESA	Engg. Science and Arts	9	24	40	19	5	21
OE	Open Elective	6	6	18	0	6	18
HM	Humanities	6	10	16	5	5	11
PSI	Project, Seminar, Internship	13	10	18	0	10	18

B. Tech VLSI Credit Distribution

TOTAL CREDITS (Including 1st year): **180**